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## **Converting Depleted Oil and Gas Wells to Potential Geothermal Resources**

Damir Zadravec, Fika Eco, Zagreb, Croatia

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### **Abstract**

The revitalization of the existing hydrocarbons wells for the purpose of testing and confirming geothermal reserves involve a series of interconnected petroleum-engineering operations. Objective of the paper is to present the development of the strategy for the conversion of the oil or gas wells, previously drilled and intended for the hydrocarbons production, into geothermal resources. To convert temporary abandoned hydrocarbons' well to serve as a part of geothermal doublet system, well site should be prepared, the well cellar refurbished, and a stable platform secured for equipment installation. This should be followed by well killing, disassembly of the Christmas tree, and installation of a blowout preventer system. After pulling the existing production or safety string and cleaning the production section open hole and cased hole, electric wireline logging should be conducted to determine reservoir properties and assess casing condition (well integrity). In the final phase of the workover, after installing a new geothermal wellhead, well stimulation and well testing equipment should be set up. During the production test, water should be produced without already pulled a small diameter gas production string, and by flowing through existing open hole, 0,178 m (7") liner and 0,244 (9 5/8") casing. Dynamic pressure continuously should be monitored using a downhole gauge. Separated gas should be flared, while water will be directed to a lagoon. After testing, produced geothermal water should be reinjected into the reservoir via the injection well. Aim of this paper is to show the engineering solutions and new business possibilities for the oil industry and help their transition into renewable energy resources. By presenting the functional requirements of the geothermal well's integrity and necessary engineering activities for safely converting of the depleted gas producer well into geothermal well, paper will bring the new and additional information's about difference between hydrocarbons and geothermal well design, well testing and later, different solutions for the produced geothermal water utilization.

**Keywords:** geothermal well, temperature gradient, well integrity, thermal energy, organic Rankine cycle

### **Introduction**

Geothermal wells are petroleum - mining objects of vital importance for the exploration and exploitation of geothermal water for energy purposes. Drilling and completion of new geothermal wells or the repurposing (converting) of existing depleted oil or gas wells enables research and testing of potential geothermal

resources. Drilling and completion of geothermal exploratory, as well as future exploitation and injection wells, provides access to the potential reservoirs and consequential interaction with the collector rocks of geothermal water (Axelsson, 2013). After drilling reservoir section of hole, electrical wire line logging measurements should be recorded across geothermal reservoirs with the aim of collecting petrophysical and reservoir data. Recorded data will serve as a base for the reservoir modeling and eventual stimulation and completion design. Finally, hydrodynamic testing of the wells should be carried out to better understand the production potential of geothermal systems. Through their interpretation, analysis and the geothermal system modeling, the preconditions for the confirmation of the geothermal water reserves are achieved, and thus the possible optimization of future geothermal water exploitation for energy purposes.

Modeling should begin with an understanding of the type of geothermal system under consideration, which can greatly assist in the identification of system utilization opportunities and the commercial viability of the project. For the development of the enhanced geothermal systems, it is not only important to use efficient equipment, but it is also important to adapt the geothermal resource to a reliable and proven technology or utilization methodology. As already emphasized, drilling and completion of geothermal wells enables access to the geothermal fluid, which contains thermal energy, and with that, the prerequisites for insight into the geothermal reservoir itself are achieved.

In general, geothermal wells are made for the purpose of research, production, injection, measurement or monitoring of key exploitation indicators of the geothermal systems performance. While performing the petroleum - mining operations in geothermal wells, presence of high temperature conditions due to increase in the temperature gradient of the rocks in depth is implied. Considering that, during the construction of new geothermal wells or the use of depleted or dry oil or gas wells as a geothermal resource, all the well construction elements (casing, tubing, submersible pumps, well head, etc.), the mud properties and cement slurries characteristics, must be designed or fixed to meet the aforementioned high temperature resistance conditions. Furthermore, the telemetry tools used while drilling of the wellbore or sidetracking, and the tools for electrical wireline logging must be temperature resistant in accordance with the expected conditions in the borehole. To ultimately ensure the greatest possible flow (mass flow rate) of geothermal water, it is important that the diameter of the well bore through the reservoir is as large as possible, which makes geothermal wells more expensive compared to oil or gas wells and dictate the appropriate drilling rigs selection and utilization. Furthermore, well integrity of geothermal wells is one of the main priorities when using "old" oil and gas wells for the purpose of obtaining geothermal water (energy).

Geothermal wells play an important role in researching the physics i.e. intrinsic properties of geothermal reservoirs, which are practically ineffective without uncovering and direct access to the reservoir, as provided by wells (Axelsson, 2013). Geothermal reservoir physics is a scientific discipline that deals with the transfer of mass and energy in geothermal systems and geothermal wells. Various modeling techniques within the dynamic system attempt to understand and quantify the flow along with the consequent changes in reservoir conditions, especially those caused by geothermal water exploitation. During the initial phase of geothermal potential research, the research relies on the analysis of geological, geophysical and geochemical data obtained from surface research. After drilling and testing the wells, and when collected data on the origin and size of the reservoir and the conditions in the geothermal system are processed, the information's obtained are used to predict the behavior of the reservoir and production wells. Based on the mentioned, the production capacity of the geothermal reservoir can be estimated, and the prediction of the behavior of the reservoir can greatly help in the management of geothermal reservoirs during future exploitation (Axelsson, 2013).

Since geothermal energy becoming an increasingly important resource worldwide amid the transition toward a low-carbon economy, particular exploratory activities already started happening at the Pannonian Basin in Republic of Croatia, which is characterized by an elevated geothermal gradient. Rich history of exploratory drilling for hydrocarbons exploration and exploitation in mentioned area, offers significant but underutilized potential for geothermal energy research.

The aim of this paper is to present an example of the conversion depleted hydrocarbons well into geothermal well. In addition, paper will present old wells integrity evaluation and revitalization process. Finally, it will show the possibilities of the geothermal energy applications. Screening criteria for the well selection as candidate for the conversion from hydrocarbons to geothermal well can be divided in two categories:

1. Well potential for the conversion - temporary abandoned depleted hydrocarbons well, located near by potential geothermal reservoir or water flooded oil reservoir with preserved inflow performance and vertical lift performance. Such kind of wells aren't permanently abandoned with squeezed cement in the reservoir and without installed permanent mechanical barriers.
2. Well potential for geothermal water production - converted hydrocarbons wells to geothermal wells with preserved well integrity. Such kind of wells can sustain long term safe geothermal water production.

The energy utilization of geothermal water can be divided into thermal and electrical applications, primarily depending on the fluid temperature. Lower-temperature waters (up to 120 °C) are mainly used for thermal purposes, especially district heating, while higher temperatures are suitable for electrical power generation. In addition to temperature, other critical factors influencing utilization include the chemical composition of the water, production sustainability, distance to consumers, and the project's techno-economic feasibility. In accordance with Croatian regulations, any investor who completes all prescribed works from the license within the selected exploration area and proves the existence of an economically viable geothermal potential may apply for an exploitation license. Upon obtaining this license, the exploration area is reclassified as an exploitation field, granting the right to permanent use of geothermal waters within the predefined time domain.

Under the decision granting the exploration license for geothermal waters in the geothermal water exploratory area at the north-west part of Croatia, the Investor (information available by request to the author), committed to carry out magneto telluric geophysical measurements, reprocess existing seismic data, and develop a geological-geophysical study. In addition, the company committed to perform well workovers, convert depleted two gas wells into geothermal wells, one serving as water producer and another one as injection well, conduct hydrodynamic measurements, as well as to interpret the obtained measurements data and analyses. After successful workover operations which will be presented in this paper, the Investor performed the well testing operations and announced commercial discovery of the geothermal water. In accordance with the Croatian law on hydrocarbons exploration and exploitation (URL1), which covers the geothermal water exploration and exploitation for energetical purposes, after commercial discovery announcement, exploitation field should be establish and prepared and approved field development project. In this case, geothermal water exploitation field will be developed by usage of two existing wells (one production and one injection) and with drilling a new five geothermal wells (three production and two injection). In addition, Binary Organic Rankine Cycle electrical power plant with all belonging pipelines and infrastructure will be constructed in purpose of electrical power generation. Granted exploitation period is 25 years and could be extended.

For this paper, nomination of the wells will be K-1 for water producer well and L-1 for the injector well. The K-1 well was originally drilled in 1969 and was revitalized as part of geothermal water exploration for future development. The well site was prepared for the installation of workover equipment and a test pit for production testing. The wellbore was revitalized, followed by hydrodynamic testing in second half of 2023. After completing the tests, the well demonstrated promising production characteristics. This paper provides a technical evaluation of the K-1 well, compares the results with regional trends, and explores the potential for commercial application. The primary objective of the well was to investigate the presumed

geothermal aquifer within Mesozoic carbonates. The trajectory of the K-1 geothermal exploratory well is vertical, with a total depth of 2430 meters.

## Geological and Geothermal Background of the exploratory area

The Pannonian Basin was formed during Miocene rifting, during which a sequence of clastic and carbonate rocks was deposited over a metamorphic basement. The exploratory area is characterized by elevated heat fluxes (70-90 mW/m<sup>2</sup>) and numerous fault systems that facilitate thermal fluid circulation (Tuschl et al., 2022). Data from offset wells indicate a geothermal gradient of 45-55 °C/km and increasing mineralization with depth. The primary reservoir horizon in K-1 well is found within Pontian and Lower Sarmatian sandstones at depth from 2200-2600 m. At 2284 m, the well penetrated Mesozoic rocks, specifically low-grade metamorphosed marly limestones and dolomitic breccias. Previous DST (Drill Stem Test) conducted in purpose of the hydrocarbon's exploration discovered low salinity water (5.26-5.85 g/L NaCl) and pressure data indicated high permeability of the reservoir rock. The water contained carbon dioxide and methane. The reservoir water temperature was between 100 and 110 °C, and production was 635 m<sup>3</sup>/day. Additional production tests were conducted within the dolomitic breccias. The reservoir is recharged by lateral inflow of cooler water from shallower aquifers and vertical inflow of deeper, warmer fluids. The density of natural thermal springs (> 60 °C) in nearby area confirms active convective processes. Fluid samples were collected at the surface and downstream of the two-phase separator. Water samples analyses were carried out in an accredited laboratory, and clay and sandstone cuttings samples were subjected to petrographic and XRD (X-ray diffraction) analysis to determine mineral composition. The presence of sodium cations and chloride anions, comprising over 80% of total dissolved salts, was confirmed. The analyses show that the produced water is sodium-bicarbonate geothermal water, with elevated levels of free carbon dioxide and dissolved hydrogen sulphide, along with methane and nitrogen. Traces of ethane and propane were also detected. Water mineralization classified it as very hard water and the concentration of free CO<sub>2</sub> in the gas phase exceeds 90%. The salinity is 5.3 g NaCl/dm<sup>3</sup>. The water causes scaling and is corrosive. (Srpak et al., 2025).

## Technological background and Technical Plan for Well Workover and Hydrodynamic Testing

The revitalization and workover of the well for the purpose of testing and confirming geothermal reserves involve a series of interconnected petroleum-engineering operations, whose efficiency depends on technical reliability, safety, and environmental acceptability. The process unfolds dynamically, in phases, depending on the petrophysical and geomechanical properties of the reservoir. According to Zarrouk and McLean (Zarrouk and McLean, 2019), several authors investigated the production of geothermal energy from abandoned petroleum wells (Bu et al., 2012; Alimonti and Soldo, 2016; Caulk and Tomac, 2017). More recently, topic was investigated by Kurevija and Macenić, for the purpose of decarbonization process by application of geothermal energy from abandoned hydrocarbons wells for district heating (Kurevija & Macenić 2022). Usually, petroleum reservoirs are flooded with water during the enhanced oil recovery stage before abandonment. These systems are effectively warm water systems, with thermal conduction as the main mode of heat transfer. These wells are already permeable since they were used for hydrocarbons production and there are no requirements for permeability enhancement. Also, the working fluid (water) should already be available in situ either from enhanced oil recovery (water flooding) or preexisting in the reservoir. However, casing integrity issues can arise when the well, originally designed for petroleum industry standards, gets converted for geothermal energy use later in its life. Therefore, proposed well design should be examined to make sure is it compatible with the geothermal drilling standards and particular geological requirements (Zarrouk and McLean, 2019).

To convert temporary abandoned gas well K-1 to geothermal, plan was to prepare the well site, refurbish the well cellar and to construct a stable and safe wellsite for equipment installation. This should be followed



by well killing, disassembly of the wellhead (Christmas tree), and installation of a blowout preventer (BOP) system. After pulling the existing safety string and cleaning the production section open hole, open-hole and cased-hole electric wireline logging should be conducted to determine petrophysical properties of the reservoir and assess casing condition and cement bond (well integrity). In the final phase of the workover, after installing a new wellhead, coiled tubing equipment and a two-phase separator should be set up. Prior to testing, the open-hole section (2430-2167 m) undergoes acid stimulation to enhance productivity. Plan was to pump 50 m<sup>3</sup> of 15% hydrochloric acid only for the scale removal purposes from the nearby borehole zone, but since well started flowing naturally after lifting operations with N<sub>2</sub>, acid treatment was postponed to the commencement of the geothermal water exploitation, then comparison between flow without and with acid stimulation and influence on well permeability will be evaluated.

During the production test, water should be produced without a production string, flowing through a open hole, 0,178 m (7") liner and 0,244 m (9 5/8") casing. Dynamic pressure continuously should be monitored using a downhole gauge (slickline). Separated gas should be flared, while water will be directed to a lagoon and transported to the L-1 wellsite for injectivity testing. After testing, excess fluid should be reinjected into the reservoir via the injection well. The power output from a doublet can be limited by the size of the well casing and the size of down-hole pump that can be installed. However, there are lots of abandoned petroleum wells globally which can potentially produce a significant amount of energy at relatively low cost when there are favourable economic conditions. (Zarrouk and McLean, 2019).

## Repurposing and preparation of Well K-1 for Hydrodynamic Testing

As per mentioned above and by following the plan to prepare well K-1 for production testing, an extensive series of petroleum-engineering operations was carried out in accordance with regulations on safety, well integrity, and production efficiency. By this case study, converting of the depleted gas well to geothermal resource will be presented. The operations began with valve inspection and well killing using a brine solution. Line tightness was verified with a pressure test of 210 bar. Once well conditions stabilized, a bidirectional safety valve was installed, the existing wellhead (Christmas tree) was dismantled, and a blowout preventer (BOP) assembly was installed. In case of revitalization of the permanently abandoned well, where there is no existing of surface wellhead and Christmas tree, civil works should be carried out to dig and clear area around old cellar, dig enough to find the casing cut and weld landing joints to allow the BOP installation. In K-1 case, after Christmas tree removal and BOP (blow out preventer) installation, the 0,073 m (2 %) safety tubing string was pulled out. To assess the integrity of casing strings and the cement sheath (well integrity), electrical wireline logging was carried out in the cased-hole section. Prior to logging, the 0,178 m (7") liner was mechanically cleaned down to 2167 m and the 0,244 m (9 %") casing to 897 m, with the simultaneous injection of chemical washes and additional viscous fluid. To verify well pressure tightness, and allow wellhead replacement, retrievable mechanical plugs were set at depths of 890 m and 2160 m. The tightness of the 0,244 m (9 %") casing and 0,178 m 7" liner was tested under 90 bar pressure for 30 minutes, with an allowable pressure drop of 10%. This was followed by mechanical cleaning of the open-hole down to total depth (2430 m) using a rotary bottom-hole assembly (0,149 m (5 %") bit, five heavy-weight drill pipe stands, and a jar), with circulation and pumping of high-viscosity pills (Marsh funnel viscosity of 100 s). Inflow was monitored during the operation, and well conditions were stabilized. After cleaning, electrical wireline logging of the open-hole section was planned. The final stage involved replacing the wellhead. After removing old adapters and flanges, new seals and the main flow control equipment were installed, a 10" API 6D Geothermal valve (rated up to 280 °C) was mounted, and the entire assembly was completed with a horizontal flow head and tested for tightness in accordance with the manufacturer's specifications. Based on above-described operations, gas to geothermal well conversion model as per K-1 sequence of the operations was developed.

## Gas to geothermal well conversion model as per K-1 sequence of the operations

### "Killing" the well, BOP installation and pulling out of hole production string

- Bull heading the well by injecting brine (salt water) with fluid density (MW - mud weight) of 1050 kg/m<sup>3</sup>.
- Nipple down Christmas tree and nipple up BOP (blow - out preventer) OD (outside diameter) 0,179 m (7 1/16") 3M.
- Pulling out of hole the tubing hanger and 33 pieces of tubing - OD (outside diameter) 0,073 m (2 7/8").

### Clean out trip inside 0,178 m (7") liner

- Assembled and run in the hole the 0,152 m (6") taper mill and clean out assembly to the 0,178 m (7") liner setting depth, while rotating and circulating. At liner shoe, pumped 2 m<sup>3</sup> of well wash and 12 m<sup>3</sup> of Hi viscous pill and circulate hole clean. Flow check and pull out of hole clean out assembly.

### Scrape 0,178 m (7") liner, set mechanical retrievable plugs and pressure test the 0,178 m (7") liner and 0,244 m (9 5/8") casing

- Assemble and run in hole the 0,178 m (7") scraper with 14 A knives to the 0,178 m (7") liner TD at of 2167,00 m.
- Reciprocate and circulate hole clean and pull out of hole 0,178 m (7") scraper assembly.
- Assemble and run in the hole the first 0,178 m (7") "C-1" retrievable mechanical plug to 2159,6 m and set the same.
- Pressure test integrity of 0,244 m (9 5/8") intermediate casing, 0,178 m (7") liner against first "C-1" plug to pressure of 90 bar for 30 minutes. Test results with AP= 4 bar, which is acceptable.
- Release and pull out of hole the setting tool to surface.
- Assembled and run in the hole the second 0,178 m (7") "C-1" retrievable mechanical plug to 858,2 m and set the same.
- Pressure test integrity of second "C-1" plug and casing to pressure of 90 bar for 30 minutes. Test results with AP= 4 bar, which is acceptable.
- Release and pull out of the hole the setting toll to surface.

**\*\*Note:** minimum two mechanical barriers are required to dismantle the BOP and wellhead.

### Nipple down the 0,179 m (7 1/16") 3M BOP and dismantle the tubing spool and casing head housing

- Lay down bell nipple and flow line
- Lay down 0,179 m (7 1/16") 3M BOP stack
- Lay down 0,179 (7 1/16") 3M × 0,346 m (13 5/8") 3M tubing spool
- Lay down 0,346 m (13 5/8") 3M × 0,346 m (13 5/8") 3M intermediate casing spool
- Inspect the top of cement behind 0,244 m (9 5/8") casing. TOC (top of cement) found - 2 m from the ground level.

## New geothermal wellhead API 3000 and BOP installation

- Replace current conventional wellhead for hydrocarbons production with new Wellhead API 3000: cut off and lay down the 0,339 m (13 3/8") surface casing string below casing head housing and weld 0,244 m (9 5/8") extension joint. Weld the surface 0,339 m (13 3/8") casing string to new casing head housing 0,339 m (13 3/8") × 0,346 m (13 5/8") 3M.
- Cooling of welds on surface 0,339 m (13 3/8") and intermediate 0,244 m (9 5/8") casing string.
- Perform NDT (Non-destructive testing) on welds - acceptable.
- Nipple up of compounds of new Wellhead API 3000:
  - Casing Head Housing: 0,339 m (13 3/8") CSG SOW bottom × 0,346 m (13 5/8") 3M RX-57 top,
  - Expansion spool: 0,346 m (13 5/8") 3M × 0,279 m (11") 3M,
  - Master valve 0,254 m (10") Ball Valve type AST2A 0,279 m (11") × 0,254 m (10") × 0,279 m (11") × API 3000.
- Nipple up the BOP stack 0,279 m (11") 5M with appropriate adaptor spool: Pressure test integrity by cup tester (0,244 m (9 5/8"), 43,5 pounds per foot) to 70 bar for 30 minutes. Test results with AP=0 bar, good test.
- Nipple up of bell nipple and flow line.

## Removal of mechanical retrievable plugs and scrape the 0,244 m (9 5/8") casing

- Run in the hole the setting tool twice and retrieve both mechanical plugs.
- Assemble and run in the hole the 0, 244 m (9 5/8") scraper (with 16 B knives) and scrap the 0,244 m (9 5/8") casing.
- Pumping and reverse circulation of 2 m<sup>3</sup> well wash, followed by 5 m<sup>3</sup> high viscosity pill (Marsh viscosity =100 sec), displace the hole with new brine MW (fluid density) of 1050 kg/m<sup>3</sup>.
- Pull out of hole with 0,244 m (9 5/8") scraper assembly.

## Inside casing electrical wire line logging (corrosion logs and cement bond logs)

- 1st Logging was performed by Multifinger Caliper56 (Mfc56) / Electromagnetic Defectoscope.
- Corrosion damage detected in intervals 900-905 m and 1035-1037 in 0,178 m (7") liner.
- From the 199 joints, there are 2 (1.01%) joints that belong to Class IV (> 50%) wear. These zones need to be seriously rechecked and considered as potential danger for pipe integrity. Sometimes such damage may occur in perforation zones and if only from single perforation it may not be so alarming. There are 63 (31.66%) joints that belong to Class III wear (25 - 50%) and could be considered as potential danger for pipe integrity. There are 32 (16.08%) joints that belong to Class II (15 - 25%) wear that are not considered critical for the pipe integrity. Most of the joints, 102 (51.26%) are not showing any significant damage and belong to Class I (< 15%) wear. The maximum joint damage/wear is Joint 7 and 11, with maximal damage of 92.1% at 1035.85 meters, and remaining wall thickness of 6.822 mm. The following figure shows casing pipe joints damage/wear class distribution histogram.
- In 2<sup>nd</sup> Logging run was performed a cement bond log, gamma ray, casing-collar locator in 0,178 m (7") liner and 0,244 m (9 5/8") casing string.
- The interpretation is based on usage of multifinger calliper MFC56 tool in combination with EMDS (Electro Magnetic Defecto-Scope). This approach enables both inside and outside casing

wear or damage detection. The inside diameter of pipes is calculated based on multifinger calliper measurement while the outside diameter of pipe is calculated by superimposing and combining thickness measurement from EMDS (Electro Magnetic Defecto-Scope) tool and interpretation of Relative Damage Distance curve, which performs evaluation whether the damage is from inside or from outside of the pipe.

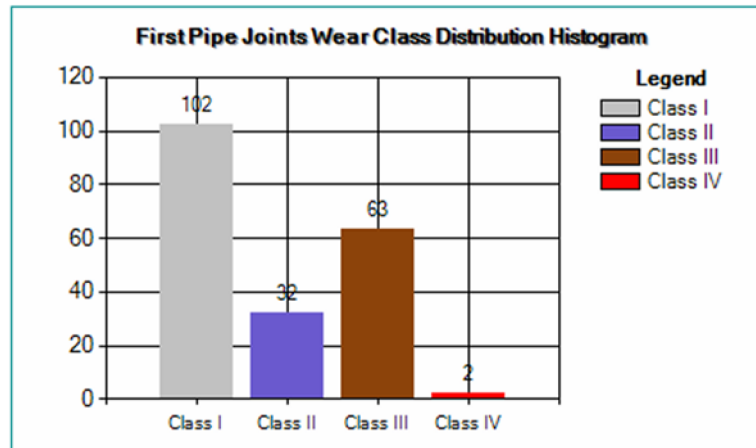


Figure 1—Joints wear Class distribution histogram

From the post logging casing wear interpretation and classification, it is evident that potential hazards for the well integrity exists in the areas which are classified as Class III and especially Class IV. Since the casing pressure tests were acceptable and good cement bond behind the casing were detected by the cement bond logs, recommendations to the operator were to continue with the operations of the preparation the well for geothermal well testing purposes. In case of extended geothermal water production, relining was recommended across the Class III area. Favourable condition for the well integrity is that only two joints of casing was classified as Class IV. To ensure the sustainable long term well integrity in the conditions of constant geothermal water production in the future, these two joins will require casing patch. However, all these well interventions activities should be verified by cost benefit analysis. Relining and casing patch will cause inside diameter reduction and will have impact on total mass flow, but on the other side will improve well integrity.

#### Clean out trip of open hole to TD (total depth)

- Assemble and run in the hole the 0,149 m (5 %") bit clean out BHA (bottom hole assembly).
- Wash down inside open hole to 2430,0 m, with a flow rate of 500-600 l/min and pressure of 50-60 bar.
- At bottom, pump 10 m<sup>3</sup> of high viscosity pill (MV=100 s) followed by 72 m<sup>3</sup> of brine (density 1,05 kg/l).
- Circulate hole clean with a flow rate of 600 l/min and pressure of 60 bar. Hazardous gases while circulation:
  - H<sub>2</sub>S=1 ppm, CO=92 ppm, CH<sub>4</sub>=4% LEL, CO<sub>2</sub>=0,25%.
- Pulled out of hole clean out assembly after circulation.



## Pressure tests the wellhead and dismantle BOP

- Pressure test integrity of 0,254 m (10") master valve (Ball Valve type AST2A 0,279 m (11") × 0,254 m (10") × 0,279 m (11") × API 3000), production elbow 0,279 m (11") × 0,079 m (3 1/8") 3M and covering tubing spool 0,079 m (3 1/8") 3M to pressure of 90 bar for 30 minutes. Test result is acceptable (DP = 0 bar).
- Nipple down and release workover rig.

Developed model is based on workover operations performed at the K-1 geothermal well and shows the necessary sequence of the operations for the converting hydrocarbons to geothermal well. It highlights the importance of proper schedule of the operations and especially requirements for well integrity evaluation by different logging techniques and pressure tests. Installation of the Geothermal wellhead with expansion spool allows the potential 0,244 m (9 5/8") casing movement due to thermal expansion and prevent entire wellhead movement or damage.

## Hydrodynamic Testing of the Well

Hydrodynamic testing of wells K-1 and L-1 was conducted to determine the petrophysical properties of the reservoir. Throughout the test, the following parameters were continuously monitored: total fluid flow rate (l/s, m<sup>3</sup>/h), wellhead and separator pressure (bar/MPa), temperature at the inlet and outlet of the separator (°C), gas-to-water ratio (GWR), gas phase flow rate (m<sup>3</sup>/h), and water flow rate (m<sup>3</sup>/h). Dynamic pressure measurements were taken using a downhole pressure gauge lowered on a slickline. Water and gas sampling was conducted in the final stage of the test to ensure representative formation fluid samples, uncontaminated by drilling mud. The separated gas was flared, while geothermal water was directed to the lagoon via a flowmeter. Maximum flow capacity at the wellhead, separator water flow (Q<sub>Water</sub>), mass flow rate, and total fluid enthalpy were also determined. Due to the potential presence of H<sub>2</sub>S and CO<sub>2</sub>, the testing was carried out under continuous supervision by a specialized service team and in full compliance with safety protocols.

## Injection Testing

Reinjection of 500 m<sup>3</sup> of fluid was performed at well L-1 via 0,073 m 2%" production tubing string, during which the wellhead pressure gradually increased as per data shown in [Table 1](#).

**Table 1—Injection rate and pressure**

| Injection rate (l/min) | Wellhead pressure (bar) | Reservoir pressure at 1785 m (bar) |
|------------------------|-------------------------|------------------------------------|
| 500                    | 13.106                  | 184.395                            |
| 750                    | 13.838                  | 185.658                            |
| 1000                   | 15.588                  | 186.683                            |
| 1500                   | 20.242                  | 186.692                            |
| 0                      | 19.219                  | 188.526                            |

The measured reservoir pressure  $P_{res}$  while injection with different injection rates was shown in [Table 1](#) was up to 19 MPa. Injection pressure as a function of the reservoir pressure gradient and surface injection pressure was 21 MPa (see [Equation 1](#)) where 2 MPa wellhead pressure was used as the measured data before injection. Injection pressure must be higher than the reservoir one which enables fluid intake.

$$P_i = P_{res} + P_{wh} = 19 + 2 = 21 \text{ MPa} \quad (1)$$

Where:

$P_i$  – surface wellhead pressure while injection, *MPa*,

$P_{res}$  – depleted gas reservoir pressure, *MPa*,

$P_{wh}$  – wellhead pressure, *MPa*,

The maximum injectivity rate depends on the reservoir pressure gradient and the installed wellhead at the surface, therefore the maximum injection pressure would not exceed fracture gradient and wellhead equipment pressure. For calculations, the maximum injectivity rate of 2160 m<sup>3</sup>/day (1500 l/min) was used. Reservoir injectivity index is calculated as follows (see Equation 2):

$$I_{ind} = \frac{I_r}{P_i - P_{res}} = \frac{2160}{21 - 19} = 1080 \frac{m^3}{day} / MPa \quad (2)$$

Where:

$I_{ind}$  – injectivity index, m<sup>3</sup>/day/MPa,

$I_r$  – maximum injectivity rate, m<sup>3</sup>/day,

$P_i$  – surface wellhead pressure while injection step rate, MPa,

$P_{res}$  – average reservoir pressure, MPa.

The injectivity index  $I_{ind}$  shows the reservoir pressure increasing by 1 MPa while pumping 1080 m<sup>3</sup> of water per day, where  $I_r$  was the maximum injection rate of 2160 m<sup>3</sup>/day. From presented data, a very high injectivity was indicated. In addition, high reservoir porosity is determined from the fast pressure drop after stop the injecting the water, with pressure drop to the initial pressure in less than 10 minutes.

## Production Parameters

More recent historical well testing data before workover and well conversion from gas to geothermal are shown in Table 2.

Table 2—Historical Well testing data (Tuchl, 2024)

| Well:                                     | L-1             | K-1             |
|-------------------------------------------|-----------------|-----------------|
| Well depth (m):                           | 2203,00         | 2430,00         |
| Bottom hole temperature (°C)              | 128,0           | 144,6           |
| Tested interval:                          | 1729,0 - 1755,0 | 2166,0 - 2430,0 |
| Temperature (°C)                          | 120,0           | 95,0            |
| Initial tested flow (m <sup>3</sup> /day) | 322,9           | 685,0           |

After performed workover operations at the K-1 well and lifting the well with Nitrogen with achieved the draw down pressure to the reservoir of  $A_p = 1,3$  MPa, well started flowing naturally, with expected productivity index. High permeability was indicated with fast pressure build up after closing the well, with pressure drop to the initial pressure in less than 5 minutes. Based on the measured temperature and pressure, the CO<sub>2</sub> saturation level was estimated at 0.8 (wt%). No free steam was detected in the wellbore during stable production, indicating sub-isothermal flow. The average totaldissolved solids (TDS) content was 57 g/L, dominated by Na-Cl and Na-HCO<sub>3</sub> facies. The total dissolved H<sub>2</sub>S content was 4.1 mg/L, while SiO<sub>2</sub> was 156 mg/L, indicating a potential risk of silicate scaling upon pressure reduction. While well testing operations, gases were separated by usage of two-phase separator with maximal working temperature of 165 °C and maximal working pressure of 2,0 MPa (20 bar). After separation gases were burned at the flare and water phase was directed to the water pits. In the process of geothermal water exploitation and geothermal field development, it is planned to reinject separated gases together with water through injection wells to the reservoir, for the purpose of the geothermal water production enhancement.

Classification based on enthalpy is critical in reservoir engineering as it defines both utilization technology and reinjection strategies. Alternative systems such as deep geothermal sources further broaden the classification framework. Based on enthalpy and temperature, K-1 can be classified as a low to medium enthalpy system (hot water system) (Kaya et al. 2011). A comparison with the similar fields in Hungary and in Slovenia reveals similar thermodynamic conditions. When compared to other wells in the Pannonian Basin, K-1 demonstrates a higher flow rate than average (15 L/s), although the temperature is slightly lower than geothermal water used in first Croatian ORC (Organic Rankine Cycle) power plant Velika Ciglena (170 °C).

## Potential Applications

The estimated thermal output is sufficient for:

1. District heating of approximately 1,500 households,
2. 5 hectares of greenhouse vegetable cultivation,
3. Binary ORC power plant with a nominal capacity of 1.2-1.5MWe (megawatts electric), assuming a conversion efficiency of 10-12% (Zarrouk and Moon, 2014).

Maximal thermal capacity and installed exergy power of the cycle was calculated in accordance with simplified equation derived from 1<sup>st</sup> and 2<sup>nd</sup> Law of thermodynamics and based on average constant mass flow of 50 l/s, geothermal water temperature of 140 °C, and output temperature from the cycle of 70 °C, both expressed in degrees Kelvin:

$$P_{ex} = \frac{Q_w \cdot C_{pg} \cdot \Delta T^2}{2T_{out}} = \frac{50 \cdot 4,2 \cdot 70^2}{2 \cdot 343} = 1,5 MW_e$$

Where:

$Q_w$  – constant flow rate: 50 l/s

$C_{pg}$  – specific heat capacity of water: 4,2 kJ/kgK.

$\Delta T$  – 70°K

$T_{out}$  – output temperature from the Organic Rankine cycle: 343 °K.

According to the physical characteristics of the rocks, the reservoir containing the existing wells belongs to the medium temperature reservoirs. The planned utilization is through a power plant with a Rankine thermodynamic cycle and an organic working fluid (ORC - Organic Rankine Cycle). The geothermal power plant for electricity generation uses geothermal water obtained through pipelines from the geothermal wells, and for heating the organic medium (n-pentane or n-butane), which evaporates in the heat exchanger and reaches the turbines in the form of steam where expansion occurs, and thermal energy is converted into mechanical energy to drive the turbines, which then rotate the generator. After passing through the turbines, the organic medium goes to the air-cooled condensers that serve as condensers and condense on them; in a liquefied state, it returns to the heat recuperators and to the preheater, and after the preheater to the heat exchanger. From the heat exchanger, where the medium evaporates again, it goes to the turbine and the process repeats in its closed cycle. The turbines powered by the organic medium are connected to a generator in which voltage is induced, and electrical energy is produced. After the geothermal water in the heat exchanger transfers its heat for the heating needs of the organic medium, the cooled water is returned to the mining facility through pipelines, which then injects that water through injection wells back into the reservoir. The entire cycle takes place in its closed process. (DiPippo, 2012). The binary principle is suitable for temperatures of 100 to 150 °C with energy utilization of 10 to 13%. The advantage of binary power plants is the closed system, where the used water is returned to the reservoir through the injection well with minimal heat loss. There are no losses of water or emissions of harmful gases during production. There is also a possibility of cascading use of energy (Kurevija and Gregurić, 2008).

## Environmental Impacts and Sustainability

The main challenge is high mineralization and scaling potential. A closed-loop system with full reinjection is recommended, along with phosphonate-based scale inhibition. The project offers significant environmental benefits, including reduced fossil fuel consumption and improved regional energy independence

## Conclusions

Precondition for application of the above-mentioned technology is the presence of the high temperature gradient at the area with existing depleted or temporary abandoned hydrocarbons wells. Either by geothermal heat transfer through Earth's crust by thermal conduction or convection, high temperature gradient is main power for a geothermal engineering models and solutions application. Consequently, further business case development can help the oil industry in feasible and financially acceptable transition to become dominant generator of the renewable energy resources. Alongside high temperature gradient, water saturated depleted oil or gas reservoirs are perfect candidates to be converted to geothermal energy resource. Gas residuals can be successfully separated at surface and reinjected in purpose of enhancement of the geothermal water production. Another key factor for success is the geothermal well's integrity and necessary engineering activities for safely converting of the depleted oil or gas producer well into geothermal well. By process of the converting the depleted gas well to geothermal resource, doublet geothermal water producer and injector K-1 / L-1, achieved stable corrected flow rate, corresponding to a thermal capacity of approximately 10 MW<sub>t</sub> (megawatts thermal) It was determined that the chemical composition of the produced geothermal water requires appropriate material protection and chemical treatment, to prevent scaling, which is important for future inhibition design. Pressure drawdown results indicate moderate to high permeability, highlighting the need for carefully designed reinjection strategies. K-1 demonstrates potential for binary power generation and/or multiple direct-use thermal applications. Further studies should focus on integration with local energy infrastructure and on long-term reservoir behaviour, reinjection optimization, especially while injecting the water into the L-1 injection well with simultaneous well testing operations at K-1 well with dynamic pressure measurement, to prove distant reservoir communication and positive effect of the water injection on doublet system for geothermal water production.

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